# marimba legio

## _travelers guide
## _for playing the asteroid belt

| 2 | 3 | 6 |
|---|---|---|
| switches · 3 ways | cv inputs | hp |

[ pitch ] press  [ pitch ] hold  › then  noon = 12 o'clock  tags = the panel drawing

physical modelling · a resonant bar as a bank of tuned modal resonators · struck, bowed or played

← back to the legio starsystem

= hold

## 01 what it is

ma_01

| | |
|---|---|
| modal | tuned resonators that ring like a bar |
| materials | tube, metal, wood |
| excite | strike or bow it with outside audio |
| in L R | excite · audio in · L feeds R if alone |
| out L R | stereo out, up to 16 V peak to peak |
| platform | a legio firmware · other firmwares |

## 02 pitch

ma_02 · enc

| | |
|---|---|
| [ enc ] turn | fine tune |
| [ enc ] press + turn | coarse tune |
| enc jack | pitch jack · –2 to +5 V |

## 10 first flight

ma_10 · the manual's tutorial

1. a pitch sequence into enc
2. triggers into trig
3. 1 + 2 at noon
4. try each s1 across octaves
5. unpatch · audio into excite · ext vs bow

## 03 decay · tone

ma_03 · knobs 1 + 2

| | |
|---|---|
| 1 [ decay ] | how long it rings |
| 2 [ tone ] | strike · soft and round → hard and bright |

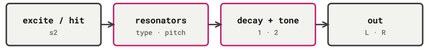
## 04 type · mode

ma_04 · s1 · s2 · top → bottom

| | |
|---|---|
| s1 type | tube · metal · wood · each likes a different register |
| s2 ext | audio strikes the resonator |
| s2 bow | audio bows the resonator |
| s2 hit | outside audio passes by · the trigger plays the voice |

## __fig.01 : signal path

in → out




as the manual describes it

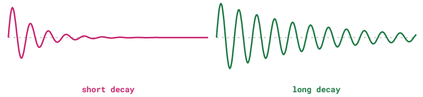
## 05 jacks

ma_05 · trig · cv · audio

| | |
|---|---|
| trig hit | trigger for the internal voice · near +1.6 V |
| in excite L R | audio in · L feeds R |
| out L R | up to 16 V peak to peak |
| cv | pitch –2 to +5 V · decay + tone 0 – 5 V |

## __fig.02 : ring out

decay knob




## 08 patch ideas

ma_08 · the manual

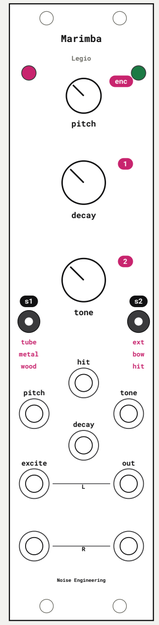
**mallets** — any · hit
pitch + triggers, decay and tone at noon

**bowed drone** — bow
a sustained sound into excite

**struck loop** — ext
a drum loop into excite strikes the bar

## __fig.03 : the panel

numbers match the legend




- enc **pitch** · fine tune
- 1 **decay** · how long it rings
- 2 **tone** · strike
- s1 **type** · tube · metal · wood
- s2 **mode** · ext · bow · hit
- trig **hit** · trigger for the internal voice
- cv **pitch** · **1** · **2** · pitch –2…+5 V · others …
- in **excite L / R** · L feeds R if alone
- out **out L / R** · stereo out

## 09 specs

ma_09

| | |
|---|---|
| size | 6 hp · 1.5 in deep |
| power | 140 mA +12 V · 22 mA –12 V |
| cv in | 0 – 5 V · pitch –2 to +5 V |
| trig | rising edge near +1.6 V |
| boot LEDs | magenta / green |